Start

Discrete weak form of step-20 like step-8

$u_h \in RT(k)$, $p_h \in DG(k)$

$$(K^{-1}u_h, v_h)_\Omega - (p_h, \nabla \cdot v_h) = -(g, v_h \cdot n)$$

$$-(\nabla \cdot u_h, q_h) = -(f, q_h)$$

$\forall\ v_h \in RT(k)$
$q_h \in DG(k)$

$$\Rightarrow \quad A(\{u_h, p_h\}, \{v_h, q_h\}) = F(\{v_h, q_h\})$$

Assume $X_h = RT(k) \times DG(k)$

Find $x_h \in X_h$ s.t.

$$A(x_h, w_h) = F(w_h) \quad \forall\ w_h \in X_h$$

where, $x_h = \begin{pmatrix} u_{h_1} \\ u_{h_2} \\ p_h \end{pmatrix}$

Let $\{\Phi_i\}$ be the basis of X_h and $\{\Phi_i\} = \{(u_i, p_i)\}$

u_i: basis of $RT(k)$, p_i: basis of $DG(k)$

$$\sum_{i,j} U_j V_i A(\Phi_j, \Phi_i) = \sum_i V_i F(\Phi_i)$$

$$\Phi_j = (u_j, p_j), \quad \Phi_i = (u_i, p_i)$$

$$A(\Phi_j, \Phi_i) = \underbrace{(K^{-1}u_j, u_i)_\Omega}_{} - (p_j, \nabla \cdot u_i)_\Omega - (\nabla \cdot u_j, p_i)_\Omega \quad \text{— (1)}$$

$$F(\Phi_i) = -(g, u_i \cdot n) - (f, p_i)$$

eg:-

$$(K^{-1}u_j, u_i) = \left(K^{-1}_{00}(\Phi_j)_0 + K^{-1}_{01}(\Phi_j)_1, (\Phi_i)_0\right) + \left(K^{-1}_{10}(\Phi_j)_0 + K^{-1}_{11}(\Phi_j)_1, (\Phi_i)_1\right) \quad \text{— (2)}$$

Solve :-

$$\begin{bmatrix} M & B \\ B^T & 0 \end{bmatrix} \begin{bmatrix} U \\ P \end{bmatrix} = \begin{bmatrix} F \\ G \end{bmatrix}$$

$$MU + BP = F$$
$$B^T U = G$$

$$U = M^{-1}(F - BP)$$

$$B^T M^{-1}(F - BP) = G$$

$$B^T M^{-1} F - B^T M^{-1} B P = G$$

(1) $S = \underbrace{B^T M^{-1} B}_{\text{Schur complement}} P = \cancel{G} \; B^T M^{-1} F - G$

Schur complement → It is sysmetric and positive definite

So, we can use CG here

(2) Solve for $U = M^{-1}(F - BP)$ using P obtained above

three steps—   ~~$B^T M^{-1} B v$~~ $S\underline{v} = B^T M^{-1} B \underline{v}$

(i) compute $\underline{w} = B\underline{v}$

(2) compute $\underline{y} = M^{-1}\underline{w}$ ~ we solve $My = w$ with CG (as symmetric and positive definite)

Finally

(3) compute $B^T\underline{y}$

In deal.ii, M, B, B^T, S, M^{-1}, S^{-1} are represented as linear operators i.e we only represent their action on vectors